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Lingua inglese

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NEW SCIENCES

- Apart from the **traditional branches** of science, such as chemistry, physics, botany, there are several more recently **established** fields.
- These include Artificial Intelligence and Data Science, Quantum Information Science, Climate Engineering and Sustainability Science – to name but a few.
- **"New Sciences"** refers to highly interdisciplinary, data-driven, and technology-fused fields that have emerged to solve complex global challenges. Unlike the specialized, isolated sciences of the past, 21st-century science relies heavily on computational power, holism, and global collaboration.

NEW SCIENCES

Artificial Intelligence & Data Science:

the development of complex computational systems capable of performing tasks that historically required human intelligence, built on unprecedented data processing.

- They are closely related but distinct fields.
- Data science analyses past and present information using statistics and machine learning to find trends and solve business problems.
- AI uses data to build systems that mimic human cognitive functions—such as learning, decision-making, and natural language processing

NEW SCIENCES

- **Artificial Intelligence (AI):** The broad field of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence, such as reasoning and learning.
- **Data Science:** An interdisciplinary field that uses scientific methods, statistics, and algorithms to extract actionable insights and knowledge from raw, structured, or unstructured data.
- **Machine Learning (ML):** A subset of AI where computers learn from historical data to identify patterns and make predictions without being explicitly programmed for that specific task.

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NEW SCIENCES

AI: two different types

weak AI (narrow AI): designed to perform a narrow task, such as facial recognition or Internet Siri search or self-driving car

- many currently existing systems that claim to use “AI” are likely operating as a weak AI focusing on a narrowly defined specific function
- although it seems to be helpful to human living, there are still some think weak AI could be dangerous because weak AI could cause disruptions in the electric grid or may damage nuclear power plants when malfunctioned

NEW SCIENCES

strong AI or artificial general intelligence (AGI): the speculative intelligence of a machine that has the capacity to understand or learn any intelligent task a human being can, thus assisting humans to unravel the confronted problem

- while narrow AI may outperform humans such as playing chess or solving equations, but its effect is still weak
- AGI could outperform humans at nearly every cognitive task

NEW SCIENCES

Different functions of AI:

1. **Automation:** what makes a system or process to function automatically
2. **Machine learning and vision:** the science of getting a computer to act through deep learning to predict and analyze, and to see through a camera, analog-to-digital conversion and digital signal processing
3. **Natural language processing:** the processing of human language by a computer program, such as spam detection and converting instantly a language to another to help humans communicate

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NEW SCIENCES

4. Robotics: a field of engineering focusing on the design and manufacturing of cyborgs, the so-called machine man. They are used to perform tasks for human's convenience or something too difficult or dangerous for humans to perform and can operate without stopping such as in assembly lines.

5. Self-driving cars: use a combination of computer vision, image recognition amid deep learning to build automated control in a vehicle.

NEW SCIENCES

Data Terminology

- **Algorithm:** A specific set of step-by-step instructions or rules used by a computer to perform calculations, solve problems, or learn from data.
- **Dataset:** A structured collection of data, typically organized in rows and columns.
- **Big Data:** Extremely large and complex datasets characterized by high volume, velocity, and variety.
- **Data Wrangling (or Cleaning):** The process of cleaning, transforming, and organizing raw data into a format ready for analysis.
- **Feature:** A measurable piece of data or attribute used in an analytical or machine learning model (e.g., age, colour, or temperature).

NEW SCIENCES

Machine Learning & AI Techniques

- **Neural Network:** Computing systems vaguely inspired by the human brain, used in deep learning to recognize complex patterns in data.
- **Deep Learning:** A specialized subfield of machine learning that uses multi-layered artificial neural networks to solve highly complex problems like image and speech recognition.
- **Generative AI (GenAI):** AI models (like Large Language Models) that generate new, original content—such as text, images, or audio—based on the data they were trained on.
- **Classification:** A supervised machine learning task used to predict discrete categories (e.g., determining if an email is "spam" or "not spam").

SCIENCE vs TECHNOLOGY

- Technology applies scientific knowledge to practical problems and also allows for new ways of doing scientific experiments. Science and technology mutually drive each other's progress.

- The main difference between science and technology is in their aims. Science has aims internal to the scientific method: the production of knowledge and understanding. Technology has aims external to its own development: technology is made to solve the practical problems of the day.

THANK YOU FOR YOUR ATTENTION!

A HISTORY OF MOBILE PHONES AND SMARTPHONES 1.

Read the first part of the article.

When was the first mobile phone invented?
The first call made from a mobile phone happened on 3 April 1973 in New York City. It was made by the engineer Marty Cooper, who worked at the telecommunications firm Motorola and the call was to a rival at his company's competitor, Bell Laboratories.

The prototype device Cooper called from was very different from the slim handsets we slip into our pockets and bags without a second thought today. Eleven years later, it finally went on public sale under the name Motorola Dynatac 8000X. In 1984, the Dynatac, known as a cell phone rather than a mobile, would cost you $3,995 - that's approximately $11,700 or £9,500 today. At the time, the average weekly wage in the UK was around £117.

For that price tag, mobile phone owners could telephone other people - but that was it. Texting, social media and camera capabilities would come (much) later. Its battery took 10 hours to charge, providing 30 minutes of use and a 15cm (6in) antenna had to be extended before making a call. It also weighed 790g (1.7lbs) - around four times as heavy as an iPhone.

What was the first generation of popular mobile phone?
In the beginning, mobile phones were largely used by businesspeople who were on the go, needed to be in contact with clients and colleagues throughout the day and - perhaps most significantly - could afford them. They used what are seen as the first generation of mobile phone, affectionately referred to as 'bricks' today due to their large size in comparison to today's models.

The second generation of mobile phone became a more regular sight in the street. These phones used the new 2G signal (which also means 'second generation') and when they arrived in the 1990s, more teenagers became mobile phone users. The ability to text was available for the first time, on handsets which were now a lot smaller and easier to handle than the Dynatac 8000X.
From 1997, some mobile phones included games (Snake was very popular!) and it was in 1999 that cameras appeared on mobiles for the first time.

Source: https://www.bbc.co.uk/bitesize/articles/z62gjfr

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A HISTORY OF MOBILE PHONES AND SMARTPHONES 2.

Read the second part of the article.

When was the internet first available on mobile phones?
The 1990s saw many developments in mobile phone technology and it was in 1996 that the internet appeared on a handset for the first time. This particular device, the 9000 Communicator, was made by the Finnish firm Nokia.

Regarded as revolutionary, the Communicator could be opened up to reveal a keyboard and screen. It even featured prominently in the 1997 film The Saint. Val Kilmer played the title hero - using his phone to get him out of a pickle whenever necessary.

When did smartphones come out?
The question: 'What was the first smartphone?' can be tricky to answer as many experts and mobile phone historians have different answers. This is because elements of what make up a smartphone, such as apps - or the ability to send emails - existed on different models at different times.

One device that is often seen as the first smartphone is the IBM Simon from 1994. It had a touchscreen, ability to send a fax, and also an email and calendar function. The BlackBerry, launched in 1999, is also seen as having smartphone capabilities. It had a keyboard instead of the traditional telephone buttons on its casing and was useful for sending and receiving emails.

The smartphone as we are more likely to know it today first appeared in January 2007, when Steve Jobs of Apple announced the iPhone to an excited crowd in San Francisco. It combined the music player, the iPod, with internet capabilities and a touch screen mobile phone. In its second version, the iPhone 3G, users were also able to download apps of their choice from an online store.

Other smartphones have followed in the wake of the iPhone. Newer, updated versions of these devices are announced on a regular basis and as technology has moved on, we can now watch live TV, films and sporting events - without too much trouble - from wherever we are in the world.

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COMMUNICATION TECHNOLOGY

Communication technology allows us to communicate in different ways. Describe some examples in 8-10 sentences.

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